

# *Putting our Heads Together: The Impact of Concussions on Player Value in Professional Sports.*

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## **I. Introduction**

Concussions are widely recognized as one of the most serious injuries players incur playing professional sports. Significant research has explored the medical side of concussions and their consequences, but the financial consequences experienced by the player are less explored. Player valuation is tied into performance metrics and contract negotiations, and there has been evidence to suggest concussions, especially multiple, decrease player performance over time. This study aims to reframe concussions as an economic issue, as well as a medical one, to explore how these players are not only affected physically negatively but also financially.

## **II. Literature Review**

Concussions are defined as traumatically induced alterations in mental status (Kelly et al., 1991). Concussions are a form of Traumatic Brain Injury (TBI), and it is estimated that there are approximately 1.6 to 3.8 million sports-related TBI's annually. However, this number cannot be definitively determined as many go undiagnosed, due to the ambiguous nature of the injury (Langlois, Rutland-Brown, & Wald, 2006). Head injuries can occur in virtually any athletic activity, but occur most often in contact sports, often resulting from direct, player-to-player contact (Kelly et al., 1991; Jildeh et al., 2022). They often occur in sports when the brain is subjected to rotational or angular acceleration

forces but are not limited to direct impacts to the head (Meehan & Bachur, 2009).

Concussion-related risks are not limited to the direct trauma itself. As athletes return to play from a concussion injury, they are most susceptible to a lower extremity injury within 90 days and 1 year of the concussion injury. Concussions can impact blood flow, neuroinflammation, slow reaction time and change patterns of one's gait, all increasing the risk of additional injuries as the athlete returns to play (Jildeh et al., 2022; Romero, 2021).

### *Diagnosis*

Concussions can be graded on a scale, and patients are placed based on symptoms exhibited by the patient. Physicians have developed different ideas about exactly what classifies each concussion but have a relative consensus (Leclerc et al., 2001).

Figure 1

Concussion Grading Systems (Wachtel, 2020)

| Grade | Cantu <sup>[11]</sup>            | Colorado <sup>[21]</sup>          | Roberts <sup>[40]</sup>          | American Academy of Neurology <sup>[42]</sup>                                                    |
|-------|----------------------------------|-----------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------|
| 0     |                                  |                                   | 'Bell ringer'; no LOC; no PTA    |                                                                                                  |
| 1     | No LOC; PTA <30 min              | No LOC; confusion without amnesia | No LOC; PTA <30 min              | No LOC; transient confusion; concussion symptoms or mental status abnormality resolve in <15 min |
| 2     | LOC <5 min; PTA >30 min and <24h | No LOC; confusion with amnesia    | LOC <5 min; PTA >30 min and <24h | No LOC; transient confusion; concussion symptoms or mental status abnormality last >15 min       |
| 3     | LOC >5 min or PTA >24h           | LOC                               | LOC >5 min or PTA >24h           | Any LOC, either brief or prolonged                                                               |

LOC = loss of consciousness; PTA = post-traumatic amnesia.

Commonly reported symptoms that are reported after sustaining a concussion can be grouped into three categories: somatic, cognitive and affective (Leclerc et al., 2001). The most common somatic symptoms are headaches and dizziness. Common cognitive issues include memory loss and difficulty concentrating and usual affective symptoms are

anxiety and disturbed sleep. The gradient of symptoms helps professionals diagnose the severity of injury, using the Concussion Grading System as seen in Figure 1 (Wachtel, 2020). Experience of symptoms changes based on the athlete and situation, making them difficult to diagnose (Meehan & Bachur, 2009; Jildeh et al., 2022; Leclerc et al., 2001).

Clinicians developed a self-administered rating questionnaire, where athletes rate the severity of symptoms of their somatic, cognitive and affective symptoms. As patients self-report symptoms each day following the injury, clinicians are able to get a sense of severity of injury and track the recovery process (Leclerc et al., 2001).

Figure 2

#### Post-Concussion Symptom Questionnaire (Wachtel, 2020)

**Table V.** Post-concussion symptoms (PCS) scale<sup>[55]</sup>

| PCS                       | Severity scale (0 = none; 6 = severe) |   |   |   |   |   |   |
|---------------------------|---------------------------------------|---|---|---|---|---|---|
| Dizziness                 | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Headache                  | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Nausea                    | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Vomiting                  | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Balance problems          | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Trouble falling asleep    | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Sleeping more than usual  | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Drowsiness                | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Sensitivity to light      | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Sensitivity to noise      | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| More emotional than usual | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Irritability              | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Sadness                   | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Nervousness               | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Numbness or tingling      | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Feeling slowed down       | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Feeling like 'in a fog'   | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Difficulty concentrating  | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Difficulty remembering    | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |
| Other                     | 0                                     | 1 | 2 | 3 | 4 | 5 | 6 |

#### Long-Term Effects

A severe outcome from concussions is Chronic Traumatic Encephalopathy (CTE), a degenerative disease linked to repeated impacts to the head that can only be diagnosed postmortem. Autopsy results have shown that the disease can impact emotion, cognitive and physical centers of the brain, and manifest in later onset symptoms (Jildeh et al., 2022). With this condition, patients generally follow a trend of poor memory and disorientation, which leads to depression, paranoia, personality changes and may include violent outbursts. This progression has the highest rate of suicide in patients, and if it progresses further, it can render patients with loss of motor skills (Saulle & Greenwald, 2012; Belson, 2012; Ward et al., 2017). While the link between injuries and development of CTE is not entirely understood, there is a correlation between trauma and number of impacts that plays a role in the development of the degenerative disease, as it has only been found in individuals with a history of repeated head trauma (Lakhan & Kirchgessner, 2012). Sport played, level of competition, duration of career, age, gender and genetic disposition have all been identified as variables that contribute to the development of CTE (Stern et al., 2011). There is no known cure, echoed by the fact that it can only be identified after death currently, but research is moving toward identification and prevention during life (Lakhan & Kirchgessner, 2012).

Former NFL player, Daniel Te'o-Nesheim was diagnosed with CTE after committing suicide, following failure to receive disability benefits from the NFL. Te'o-Nesheim's personality changed drastically in his last few years of life, and in his brain autopsy, neuropathologist Ann McKee found the aggressive signs of CTE, despite being only 30 years old (Belson, 2018).

## *Concussions in Professional Sports*

Concussions have been shown to negatively affect a variety of performance metrics in professional athletes, in both cognitive health and performance (Peterson et al., 2020). There are extensive studies about long-term impacts, but the short-term is still quite ambiguous (Navarro et al., 2018). Navarro et. al.'s study in the Journal of Neurotrauma is one of few to assess the decline in performance following a concussion in professional athletes. Paul Kariya is a stark example of decreased performance after concussions. His symptoms were so severe, he was forced into an early retirement. The former All-Star and a leading scorer in the NHL, Kariya is just one athlete whose career was derailed due to brain injuries (Oland, 2017).

As concussion research evolves, each league in professional sports has developed a protocol for treating and preventing head injuries. Protocols were put in place to standardize recovery after concussions and to prevent premature return to activity following the injury (Gurarie, 2024). The National Football League (NFL) works in accordance with the NFL's Head, Neck and Spine Committee and NFL Players Association, to define the most current and optimal care for concussions and implement it each season (NFL, 2023). Prior to athletic seasons, athletes are required to take a baseline concussion assessment. Upon injury, players must complete a concussion assessment, and the play is reviewed by athletic trainers and concussion spotters. A neurological examination is performed and compared against the player's preseason baseline scores. An athlete diagnosed with a concussion is not allowed to return to play until cleared by an independent neurologist (Jacobi et al., 2023). The player must complete the 5 steps of the

Return-To-Participation (RTP) Protocol designed by the NFL. Steps include symptom limited activity, aerobic exercise, football-specific exercise, club-based non-contact training drills and finally full football activity/clearance. As the athlete demonstrates the ability to achieve each level, the player-patient is granted the ability to perform the next step (NFL, 2017; Jacobi et al., 2023). Similarly to the NFL, the National Hockey League (NHL), Major League Baseball (MLB) and National Basketball Association (NBA) have all developed their own sport-specific concussion protocol and return to play steps (Beaver, 2018; Peterson et al., 2020; McGregor, 2024). The NFL's process is more rigid compared to the others, with cases being complicated due to the sport involving increased amounts of contact that could lead to further injury. Baseline testing is used and RTP steps include various neurological examinations with trained professionals (McGregor, 2022; Deubert, 2017). Tests allow for more data to be collected about how a player's brain functions before and after the brain injury, helping medical professionals best understand the effects of concussions on professional athletes.

### *Stigma*

Despite the research on the topic and effort put into preventing and treating concussions, professionals still seek to absolve their sport of the blame for the long-term effects of concussions. NHL Commissioner Gary Bettman has continually denied the link between CTE and concussions, blaming the media for fanning the flames about long-term effects of head injuries (Branch, 2016). This came in response to a lawsuit filed by over 300 former players, who argued that the NHL failed to protect players from head injuries or warn them of the risks involved (Whyno, 2019). In his written response, Bettman stated

that “The relationship between concussions and the asserted clinical symptoms of C.T.E. remains unknown” and “This, sadly, is precisely the type of tragedy that can result when plaintiffs' lawyers and their media consultants jump ahead of the medical community and assert, without reliable scientific support, that there is a causal link between concussions and C.T.E.” (Branch, 2016). The NHL settled the lawsuit for \$18.9 million, granting \$22,000 for each player and provisions for testing, insinuating the league did have a part to play in former player’s conditions (Whyno, 2019). Several medical professionals disagree with Bettman, including Dr. Charles Tator, who posits that there is a conclusive link and enough evidence to declare a causal relationship between concussions and development of CTE (Strong, 2019).

My study aims to help fill this gap in understanding about concussions, to have a similar effect and give a data driven estimate of the impact of concussions on player performance by measuring the decrease in metrics following a concussion in professional athletes and contextualizing results in terms of player market value. By modeling the patterns in players’ performance data, we can quantify how detrimental concussions are. (Navarro et al., 2018).

### **III. Methodology**

As aforementioned, my study aims to quantify the relationship between concussions and player market value changes in professional sports. Using each player’s cap hit (contracted salary per season under the salary cap), a matched-pair observation design was employed to determine if there was a significant difference between

fluctuations in player value between players who suffered a concussion and their counterparts who did not.

My dataset was split into two groups of study; the treatment and the control. The treatment group contained 250 who sustained at least one diagnosed concussion and missed some amount of time due to injury. The control group consisted of 250 NHL players who had no history of concussions reported. Each player's yearly cap hit, games played, goals, assists, total points and penalty minutes were recorded, in addition to the concussion information. The first recorded concussion served as the event anchor point, and each player's data was manipulated in R to determine three key time points: Before; one year prior to the concussion, Concussed; the year of the first concussion, and After; the year following the concussion or the next season the player's cap hit changed, indicating a new contract was negotiated. For the control group, a random player was selected to serve as a benchmark for each concussed player. The matching control player was required to have a similar season history and needed cap hit data available for all three of the same time periods as the concussed player.

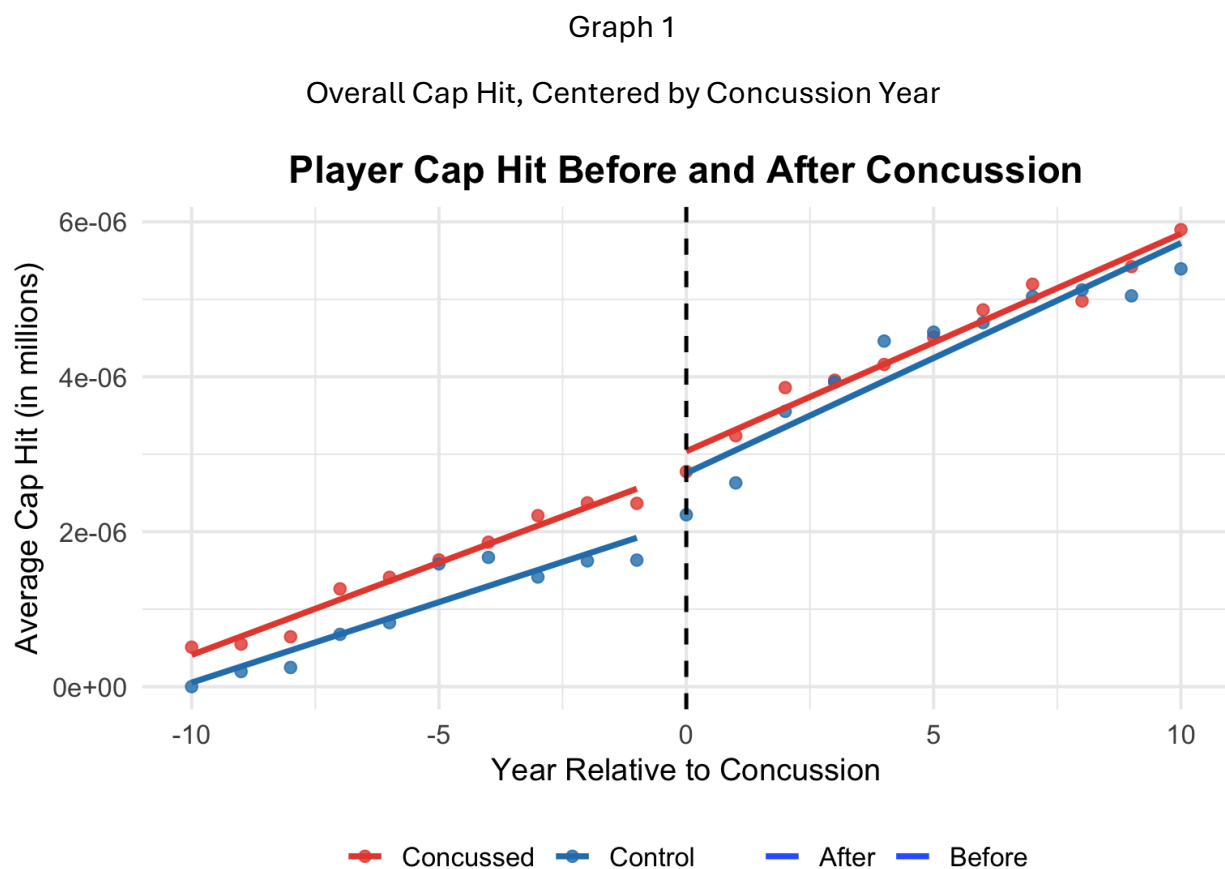
#### **IV. Analysis**

Typically, an NHL player starts on a base-level entry salary, around \$1 million per year. After the initial contract expires, players negotiate a new contract that the player's agent and team agree upon. The contract reflects the overall worth of the player to the organization, based on their overall performance, intangible traits they bring to the team, and market trends. Since the introduction of the salary cap in 2005, player contracts have



increased significantly. Teams were given a floor and ceiling for the amount they can pay their players each season, and at the point of implementation, the salary cap was \$39 million per team, and that has since increased to \$88 million. Over the course of a player's career, you tend to find a plateau in salary, with a peak associated with a player's prime and an eventual taper off with age and declining performance.

After matching and centering each player, the salary averages were graphed around Year 0, with anything prior labeled as a negative and anything following included as a positive. The data was manipulated this way to aid in visual comparison between the control versus treatment groups.



This graph clearly illustrates how average player cap hit changed over time for both

concussed and control groups, with a clear break at the year of concussion, year zero. As expected, salaries rise over time, reflecting the overall salary inflation and career progression that is typical in an NHL career. Concussed players started with a higher average salary and had a steeper slope of  $2.38\text{e-}07$ , compared to  $2.08\text{e-}07$ . The difference in slope suggests that concussed players had a higher trajectory in the years leading up to their injury, when compared against their peers who did not have a concussion during their career. After the instance of year zero, the trend reverses. The control group's slope increases to  $2.97\text{e-}07$ , while the concussed groups' slope increases modestly to  $2.81\text{e-}07$ . Both groups see an increase, where the control increased by almost a point and the concussed group by about a half point. The difference indicates a diminished growth rate for concussed players following the injury.

While the minor difference between the groups may be initially reassuring – as it suggests that teams are still investing in previously injured players and players are not losing financially due to concussions – looking deeper, this could instead be an indicator that organizations may be undervaluing the long-term impact of concussions when negotiating player value.

Table 1

Player Statistics by Cap Hit

| Summary Statistics by Group and Time Period |        |          |              |                     |              |                 |        |            |
|---------------------------------------------|--------|----------|--------------|---------------------|--------------|-----------------|--------|------------|
| Group                                       | Period | Total GP | Total Points | Total Cap Hit (\$M) | Cap Hit / GP | Cap Hit / Point | Avg GP | Avg Points |
| Concussed                                   | After  | 49,791   | 24,276       | 3,910.750           | 0.079        | 0.161           | 54.1   | 26.4       |
| Concussed                                   | Before | 27,647   | 12,800       | 995.687             | 0.036        | 0.078           | 53.7   | 24.9       |
| Control                                     | After  | 84,055   | 42,068       | 6,024.045           | 0.072        | 0.143           | 59.0   | 29.5       |
| Control                                     | Before | 18,792   | 9,268        | 533.376             | 0.028        | 0.058           | 52.3   | 25.8       |

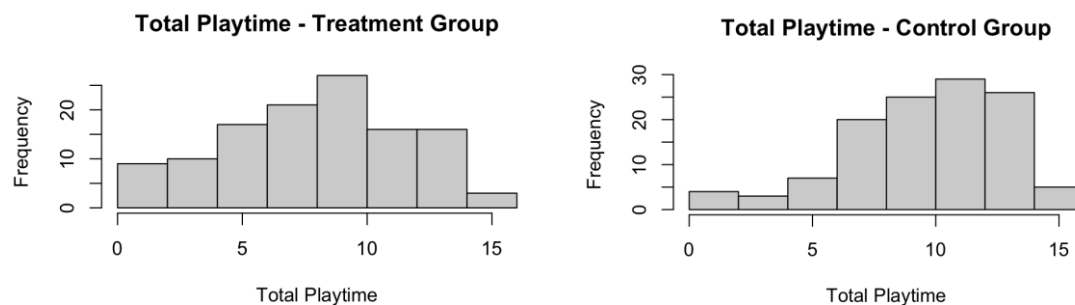
While Graph 1 visualizes the average trend in salary over time, Table 1 provides further quantified evidence to support the trend. Concussed and control players have similar point production averages before the injury, but concussed players consistently cost more per game and per point. After the concussion, the injured group costs \$161k per point and \$79k for each game despite, compared to \$143k and \$72k respectively for the control group. Despite the elevated costs, concussed players do not show the improved point production to support their cost. The control players' production increases by 3.7 points on average, compared to 1.5 for concussed players. The increased price tag can also be attributed to the fact that they play less games per season after sustaining the injury. This, combined with the lack of increase in production, indicates that these marginal gaps suggest that teams continue to pay players based on past performance, rather than working with shorter team contracts and evaluating player value on a consistent basis. It may also indicate that teams are absorbing the hidden costs of injury, helping to inflate the cost of their contracts over time.

This may seem counterintuitive to encourage organizations to pay concussed players less, but this should not be interpreted as a punishment. Instead, in a moral effort

to support their players, organizations should consciously reduce injured players workloads to aid their recovery and protect their future health. While this will decrease player value slightly in the short term, it allows them to fully recover and compete at an NHL level of play for a longer period of time.

Graph 2 and 3

### Length of Career by Group



The data supports this suggestion, with the concussed group averaging 8.34 seasons played, in comparison to 10.03 seasons for the non-concussed group. This difference was confirmed to be statistically significant in a two-sample t-test ( $t = -3.73$ ,  $p < 0.001$ ), and the two-season gap in career length indicates a quantified, long-term impact of concussions. This supports the idea that organizations should reframe the structure and contents of contracts following concussions, because this lowers the overall return on investment for the team. If a player is forced to retire due to an injury, their contract is paid out in full, and while the team is given some cap relief, they lose the overall investment as the player can no longer contribute on the ice. By taking proactive steps, such as incorporating

performance-based incentives and shorter contract terms post-concussion, teams can support the player's recovery, while also positioning the team for long-term success.

Together, this analysis shows that while players do not suffer immediate financial penalties due to concussions, their long-term health and career have an increased risk of being negatively affected. Subtle discrepancies in cap hit and career length expose another invisible impact of concussions, felt by both players and organizations. While the current system may not be disadvantaging players intentionally, organizations current practices suggest a broader lack of structural awareness of the post-concussion NHL player. By better understanding these injuries and post-concussion trajectories, organizations will be able to better balance player valuation with ethical considerations while matching the physical demands of the sport.

## **V. Conclusion**

While widely recognized for their medical implications, concussions in professional sports also carry significant economic consequences. This study reframes concussions as a catalyst for economic change in a player's career, illustrating a link between cerebral injuries and overall player performance, contract value and career trajectory. The findings of this study emphasize the importance of treating concussions as physical injuries, with long-term ramifications on one's personal and professional life. By acknowledging these invisible, yet quantifiable impacts of the injuries, organizations can take more informed steps to protect their athletes, better insuring both parties' financial and future stability.

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